



FAT EXAM SET -5

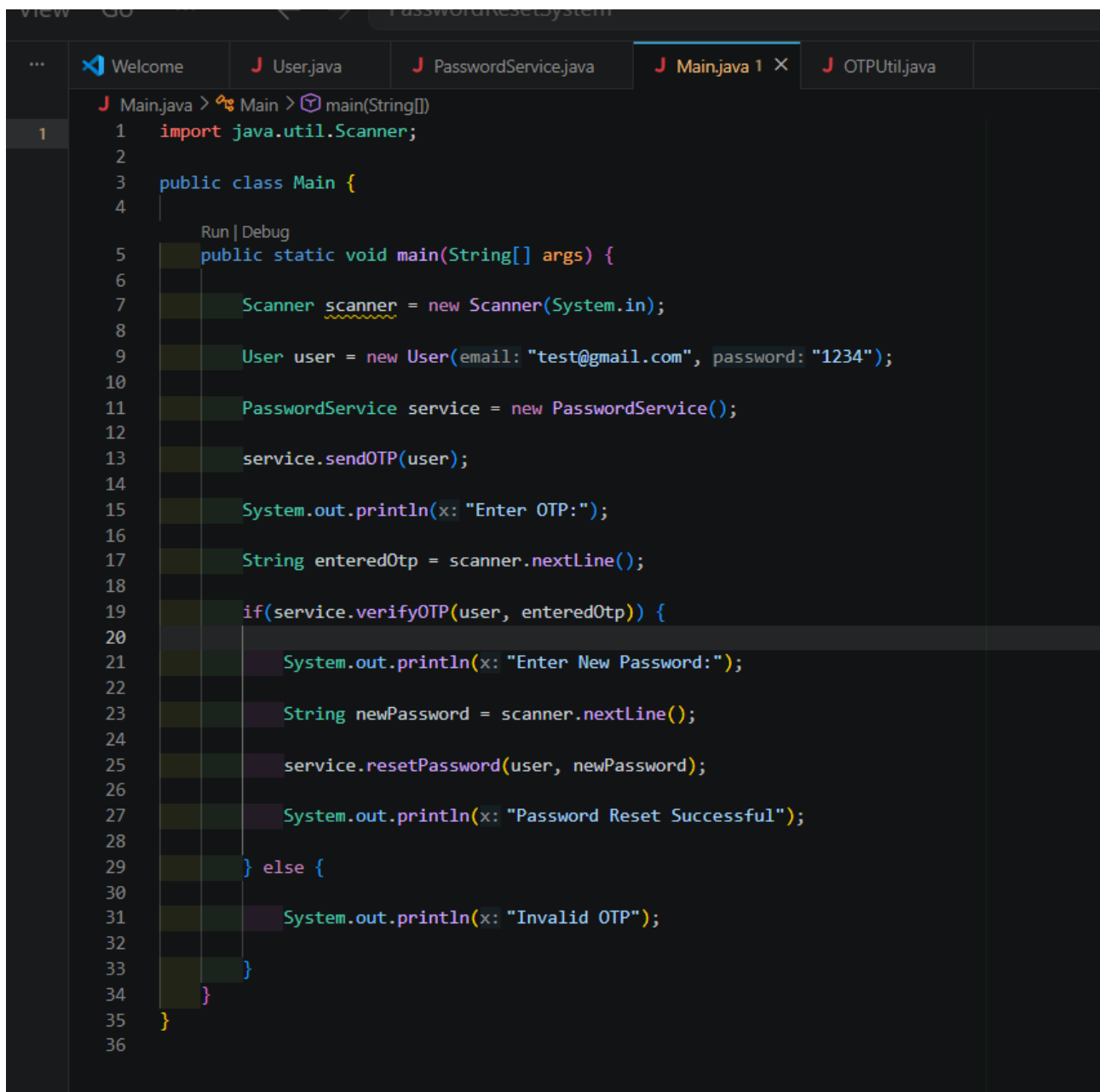
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COURSE : AGILE DEVELOPMENT AND DEVOPS PROCESS LAB

FACULTY : BALAGANESH .N

1.DEVELOP RESET LOGIN WITH OTP VERIFICATION :



The screenshot shows an IDE with a dark theme. The top bar displays several tabs: 'Welcome', 'User.java', 'PasswordService.java', 'Main.java 1 X', and 'OTPUtil.java'. The 'Main.java' tab is active, showing the following Java code:

```
1  import java.util.Scanner;
2
3  public class Main {
4
5      Run | Debug
6      public static void main(String[] args) {
7
8          Scanner scanner = new Scanner(System.in);
9
10         User user = new User(email: "test@gmail.com", password: "1234");
11
12         PasswordService service = new PasswordService();
13
14         service.sendOTP(user);
15
16         System.out.println(x: "Enter OTP:");
17
18         String enteredOtp = scanner.nextLine();
19
20         if(service.verifyOTP(user, enteredOtp)) {
21
22             System.out.println(x: "Enter New Password:");
23
24             String newPassword = scanner.nextLine();
25
26             service.resetPassword(user, newPassword);
27
28             System.out.println(x: "Password Reset Successful");
29
30         } else {
31
32             System.out.println(x: "Invalid OTP");
33
34         }
35     }
36 }
```

```
view Go ... PasswordResetSystem

Welcome User.java PasswordService.java Main.java 1 OTPUtil.java

OTPUtil.java > ...
1 import java.util.Random;
2
3 public class OTPUtil {
4
5     public static String generateOTP() {
6
7         Random random = new Random();
8         int otp = 100000 + random.nextInt(bound: 900000);
9
10        return String.valueOf(otp);
11    }
12 }
13
```

OUTPUT :

```
13     return email;
14 }
15

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER Run: Main

C:\Users\NITHYA S\Desktop>PasswordResetSystem>
C:\Users\NITHYA S\Desktop>PasswordResetSystem>
C:\Users\NITHYA S\Desktop>PasswordResetSystem>
C:\Users\NITHYA S\Desktop>PasswordResetSystem> c: && cd "c:\Users\NITHYA S\Desktop>PasswordResetSystem" && cmd /C ""C:\Users\NITHYA S\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\latest\bin\java.exe" --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\NITHYA S\AppData\Roaming\Code\User\workspaceStorage\7075c55857ec9982fb53aa9c94039c95\redhat.java\jdt_ws>PasswordResetSystem_32e2b32a\bin Main ""
OTP Sent: 896697
Enter OTP:
896697
Enter New Password:
234567
Password Reset Successful

c:\Users\NITHYA S\Desktop>PasswordResetSystem>
```

```
view Go ... PasswordResetSystem

Welcome User.java PasswordService.java X Main.java 1 OTPUtil.java

J PasswordService.java >...
1 public class PasswordService {
2
3     public String sendOTP(User user) {
4
5         String otp = OTPUtil.generateOTP();
6         user.setOtp(otp);
7
8         System.out.println("OTP Sent: " + otp);
9
10        return otp;
11    }
12
13    public boolean verifyOTP(User user, String enteredOtp) {
14
15        return user.getOtp().equals(enteredOtp);
16    }
17
18    public void resetPassword(User user, String newPassword) {
19
20        user.setPassword(newPassword);
21    }
22 }
23
24
```

2.WRITE JUNIT FOR INVALID RESET ATTEMPTS :

File Edit Selection View Go ... PasswordResetSystem

EXPLORER

PASSWORDRESETSYSTEM

> lib

J Main.class

J Main.java

J OTPUtil.class

J OTPUtil.java

J PaawordResetService

J PasswordResetTest

J PasswordService.class

J PasswordService.java

J User.class

J User.java

Welcome

J User.java

J PasswordResetTest X

J PaawordResetService

J PasswordService.class

junit-platform-console-standalone

J PasswordResetTest

1 import org.junit.jupiter.api.Test;

2 import static org.junit.jupiter.api.Assertions.*;

3

4 public class PasswordResetTest {

5

6 PasswordResetService service = new PasswordResetService();

7

8 @Test

9 void testInvalidOTP() {

10 boolean result = service.resetPassword("user1", "1234", "newPass");

11

12 assertFalse(result);

13 }

14

15 @Test

16 void testEmptyPassword() {

17 boolean result = service.resetPassword("user1", "9999", "");

18

19 assertFalse(result);

20 }

21

22 @Test

23 void testNullUser() {

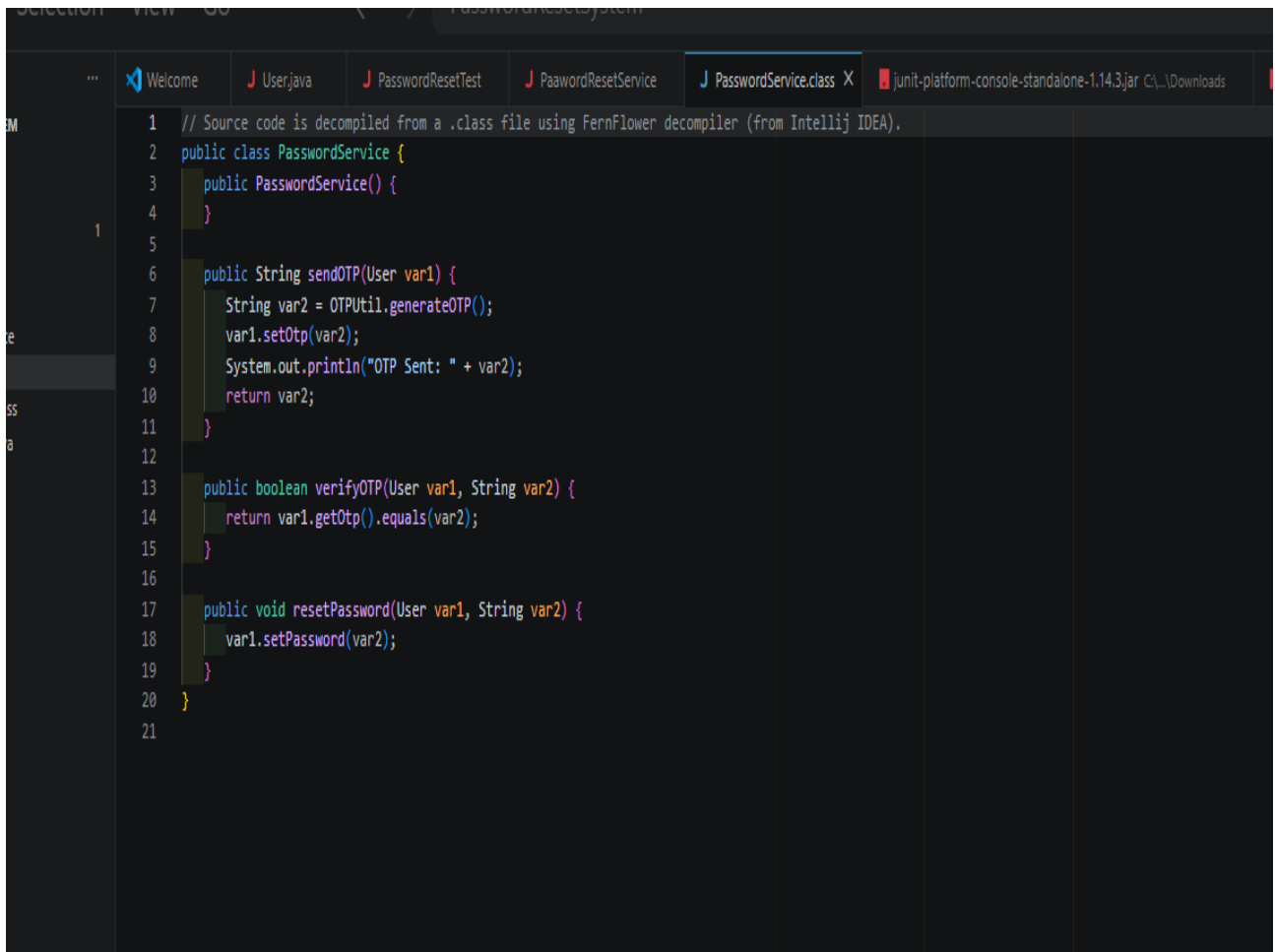
24 boolean result = service.resetPassword(null, "9999", "newPass");

25

26 assertFalse(result);

27 }

28 }



```
1 // Source code is decompiled from a .class file using FernFlower decompiler (from IntelliJ IDEA).
2 public class PasswordService {
3     public PasswordService() {
4     }
5
6     public String sendOTP(User var1) {
7         String var2 = OTPUtil.generateOTP();
8         var1.setOtp(var2);
9         System.out.println("OTP Sent: " + var2);
10        return var2;
11    }
12
13    public boolean verifyOTP(User var1, String var2) {
14        return var1.getOtp().equals(var2);
15    }
16
17    public void resetPassword(User var1, String var2) {
18        var1.setPassword(var2);
19    }
20 }
21
```

3. GITHUB

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin YOUR_REPO_LINK
git push -u origin main
```

ci.yml Code

name: Java CI

on:

push:

branches: [main]

jobs:

build:

runs-on: ubuntu-latest

steps:

- uses: actions/checkout@v3

- name: Set up JDK

uses: actions/setup-java@v3

with:

java-version: '17'

- name: Build with Maven

run: mvn clean install

```
:\\Users\\NITHYA S\\Desktop\\demo>mvn clean
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:assignment-system >-----
[INFO] Building assignment-system 0.0.1-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ assignment-system ---
[INFO] Deleting C:\\Users\\NITHYA S\\Desktop\\demo\\target
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time:  1.843 s
[INFO] Finished at: 2026-04-12T22:13:37+05:30
[INFO] -----
```

Jenkins

exam-pipeline

Configuration

Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/NITHISHVAIBHAV08/EXAM-EVALUATION-SYSTEM

Credentials

none

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

main

+ Add Branch

Repository browser

(Auto)

Save

Apply

guration

Git

Repositories

Repository URL

https://github.com/NITHISHVAIBHAV08/EXAM-EVALUATION-SYSTEM

Credentials

none

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/main/

+ Add Branch

Repository browser

(Auto)

Additional Behaviours

+ Add

Script Path

Jenkinsfile



Jenkins / exam-pipeline

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax



exam-pipeline

Permalinks

- Last build (#12), 1 min 14 sec ago
- Last stable build (#12), 1 min 14 sec ago
- Last successful build (#12), 1 min 14 sec ago
- Last failed build (#11), 5 min 46 sec ago
- Last unsuccessful build (#11), 5 min 46 sec ago
- Last completed build (#12), 1 min 14 sec ago

Builds

Filter

Today

#12 3:31 PM



exam-pipeline / #12 Console Output

Status

Changes

Console Output

Edit Build Information

Timings

Git Build Data

Pipeline Overview

Thread Dump

Resume

Replay

Pipeline Steps

Workspaces

Previous Build

Console Output

Download Copy View as plain text

```
Started by user Watson Vachey S
Obtained Jenkinsfile from git https://github.com/NETCHANDHARMS/EXAM-EVALUATION-SYSTEM
[Pipeline] start of pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\jenkins\workspace\exam-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (declarative) checkout SCM
[Pipeline] checkout
selected git installation does not exist, using default
The recommended git tool is: NONE
using credential gitlab-creds
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\jenkins\workspace\exam-pipeline\git # timeout=30
Retrieving changes from the remote git repository
> git.exe config remote.origin.url https://github.com/NETCHANDHARMS/EXAM-EVALUATION-SYSTEM # timeout=30
Fetching upstream changes from https://github.com/NETCHANDHARMS/EXAM-EVALUATION-SYSTEM
> git.exe --version # timeout=30
> git --version # 'git version 2.35.1.virtuous.1'
using GIT_ASKPASS to set credentials gitlab Access
> git.exe fetch --tags --force --progress -- https://github.com/NETCHANDHARMS/EXAM-EVALUATION-SYSTEM +refs/heads/*:refs/remotes/origin/* # timeout=30
> git.exe rev-parse 'origin/main^{commit}' # timeout=30
Checking out revision 8d9f76e5e8b3b9c0f6d5e8f8a1a4b5 (origin/main)
> git.exe config core.sparsecheckout # timeout=30
> git.exe checkout -f 8d9f76e5e8b3b9c0f6d5e8f8a1a4b5 # timeout=30
Commit message: "Update Jenkinsfile"
> git.exe rev-list --no-walk 8d9f76e5e8b3b9c0f6d5e8f8a1a4b5 # timeout=30
[Pipeline] }
[Pipeline] // stage
[Pipeline] adhocline
[Pipeline] {
[Pipeline] adhocline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (declarative) tool install
[Pipeline] tool
[Pipeline] environmentTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] adhocline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone)
```

```

Jenkins example-pipeline #12 Console Output

# exporting attestation manifest sha256:4c5889b76360d355f76362c1140110112c1388018c0870a7773204697c 0.0s done
# exporting manifest list sha256:7041a2122a6c13e1ee31a8e75d6e9a4c22ac4eb7c0ef7a230e7309a 0.0s done
# saving to docker.io/library/example-app:latest done
# unpacking to docker.io/library/example-app:latest 0.0s done
# DONE 0.2s
[Pipeline] }
[Pipeline] // without
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Docker classa optional))
[Pipeline] tool
[Pipeline] envvarsForTool
[Pipeline] without
[Pipeline] {
[Pipeline] bat

C:\ProgramData\jenkins\jenkins\workspace\example-pipeline>docker rm -f example-container || exit 0
Error response from daemon: No such container: example-container
[Pipeline] }
[Pipeline] // without
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Docker run (optional test))
[Pipeline] tool
[Pipeline] envvarsForTool
[Pipeline] without
[Pipeline] {
[Pipeline] bat

C:\ProgramData\jenkins\jenkins\workspace\example-pipeline>docker run -d -p 8080:8080 --name example-container example-app:
9b34b946c7c24b06754b0f54b0d03f0e4f9cc60017c30bc690c9377c
[Pipeline] }
[Pipeline] // without
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Check Kubernetes Connection)
[Pipeline] tool
[Pipeline] envvarsForTool
[Pipeline] without
[Pipeline] {
[Pipeline] bat

```

```

Jenkins example-pipeline #12 Console Output

[Pipeline] envvarsForTool
[Pipeline] without
[Pipeline] {
[Pipeline] bat

C:\ProgramData\jenkins\jenkins\workspace\example-pipeline>kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
example-control-plane Ready    control-plane   57m   v1.24.2
[Pipeline] // without
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] tool
[Pipeline] envvarsForTool
[Pipeline] without
[Pipeline] {
[Pipeline] bat

C:\ProgramData\jenkins\jenkins\workspace\example-pipeline>kubectl apply -f deployment.yaml
deployment.apps/example-app unchanged
[Pipeline] }
[Pipeline] // without
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declaration: post without)
[Pipeline] echo
# SUCCESS: PIPELINE SUCCESS 07h
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // without
[Pipeline] }
[Pipeline] // without
[Pipeline] }
[Pipeline] // without
[Pipeline] // without
[Pipeline] // without
[Pipeline] // without
[Pipeline] // end of Pipeline
Finished: SUCCESS

```

4. DEVELOP SERVICES ON DOCKER AND KUBERNETES :

Dockerfile Code

```

FROM openjdk:17
COPY target/password-reset.jar app.jar
ENTRYPOINT ["java", "-jar", "/app.jar"]

```

deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: password-reset

spec:

replicas: 2

selector:

matchLabels:

app: password-reset

template:

metadata:

labels:

app: password-reset

spec:

containers:

- name: password-reset

image: username/password-reset

ports:

- containerPort: 8080

service.yaml

apiVersion: v1

```
kind: Service

metadata:
  name: password-reset-service

spec:
  type: NodePort
  selector:
    app: password-reset

ports:
  - port: 8080
    targetPort: 8080
    nodePort: 30007
```

